

SIMATS ENGINEERING
CO2_ASSESSMENT TOOL3_Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I Shri Shanmuga Priya K 192324035 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge

servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

Introduction

E-commerce companies often experience sudden increases in website traffic during major shopping events such as Black Friday, Diwali, Christmas, or New Year sales. During these periods, thousands or even millions of customers may access the website simultaneously to browse products, place orders, and make payments. If the system is not designed to handle this increased demand, the website may become slow or even unavailable. Cloud computing addresses these challenges through auto-scaling, load balancing, and monitoring services, ensuring that applications remain responsive and highly available.

Cloud Auto-Scaling and Load Balancing

- Auto-scaling is a cloud feature that automatically adjusts the number of computing resources according to the workload. When traffic increases, new virtual machines or containers are launched automatically. Similarly, when traffic decreases after the sale, unnecessary resources are removed, reducing operational costs. This elasticity allows businesses to handle sudden traffic spikes without manual intervention.
- A load balancer works together with auto-scaling by distributing incoming user requests evenly across all available servers. Instead of sending all requests to one server, it ensures that each server handles only a portion of the workload. If one server becomes unhealthy or fails, the load balancer automatically redirects traffic to other healthy servers, preventing service interruption.

The combined working process is as follows:

- Customers send requests to the website.
- The load balancer distributes requests among multiple servers.
- Monitoring tools continuously check CPU usage, memory, and response time.
- If demand increases beyond a predefined threshold, auto-scaling launches additional servers.
- When demand decreases, extra servers are automatically removed.
- This process ensures that resources are always available according to user demand.

Reducing Downtime and Improving User Experience

One of the major advantages of auto-scaling and load balancing is the reduction of downtime. During high traffic, additional resources prevent servers from becoming overloaded. Even if one server fails unexpectedly, the load balancer redirects traffic to other healthy servers, allowing users to continue accessing the application without interruption. This approach significantly improves user experience by providing:

- Faster website loading time. Smooth product browsing.
- Quick checkout and payment processing.
- Reduced waiting time. Minimal buffering or delays.
- Higher customer satisfaction.

Role of Monitoring Tools

Monitoring tools are essential for maintaining application performance. They continuously collect information about system health and resource usage. These tools monitor important parameters such as:

- CPU utilization.
- Memory usage.
- Network traffic.
- Disk performance.
- Application response time.
- Number of user requests.
- Error rates.

If any metric crosses a predefined threshold, monitoring services automatically trigger auto-scaling or send alerts to administrators. Common cloud monitoring tools include:

- Amazon CloudWatch
- Azure Monitor
- Google Cloud Monitoring
- Prometheus
- Grafana

These tools also generate dashboards and reports that help administrators identify performance bottlenecks and optimize resource usage.

Real-World E-Commerce Scenario

A good example is Amazon's Prime Day Sale or Flipkart's Big Billion Days. During these sales, millions of customers visit the websites within a short period. Auto-scaling automatically launches additional application servers to handle the increased demand. Load balancers distribute requests among these servers, ensuring that no server becomes overloaded. Monitoring tools continuously observe system performance and trigger scaling actions whenever required. As a result, customers experience smooth browsing, quick payments, and uninterrupted shopping despite massive traffic.

Advantages

The use of auto-scaling and load balancing provides several benefits:

- High availability of applications.
- Efficient resource utilization.
- Reduced downtime.
- Better application performance.
- Improved customer experience.
- Automatic handling of traffic spikes.
- Lower operational costs through dynamic resource allocation.

Conclusion

Cloud auto-scaling, load balancing, and monitoring form the foundation of a highly available cloud infrastructure. Together, they enable organizations to handle unpredictable traffic efficiently, minimize downtime, optimize costs, and provide a seamless user experience during peak business periods.

2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

Introduction

Every organization generates valuable business data, including customer records, financial information, application databases, and documents. Protecting this data against accidental deletion, cyberattacks, hardware failures, and natural disasters is essential. Cloud storage provides a secure, reliable, and scalable platform for storing backup data while ensuring high durability, redundancy, and easy recovery.

Cloud Storage, Durability, and Data Redundancy

Cloud storage services are designed to provide extremely high durability by storing multiple copies of data across different storage devices and data centers. Instead of storing data on a single physical server, cloud providers automatically replicate data across multiple locations. This ensures that if one server or storage device fails, another copy remains available.

For example, cloud services such as Amazon S3, Microsoft Azure Blob Storage, and Google Cloud Storage automatically replicate data across multiple availability zones. These services continuously monitor storage devices and replace damaged copies whenever necessary, ensuring long-term data protection.

Cloud storage also supports:

- Automatic backup scheduling.
- Incremental backups.
- Cross-region replication.
- Object versioning.
- Snapshot creation.

These features significantly improve data reliability and availability.

Encryption and Access Control

Since backup data often contains confidential information, strong security measures must be implemented.

Data stored in the cloud should be encrypted using AES-256 encryption, making it unreadable without the correct encryption key. During transmission between users and cloud servers, TLS/SSL encryption protects data from interception.

Access to backup data should be controlled using Identity and Access Management (IAM) policies. Users should receive permissions based on their job responsibilities through Role-Based Access Control (RBAC). Additional security can be achieved using:

- Multi-Factor Authentication (MFA).
- Strong password policies.
- Audit logging.
- Key Management Services (KMS).

These measures ensure that only authorized personnel can access or restore backup data.

Cost Advantages Compared to Traditional Storage

Traditional backup systems require organizations to purchase expensive storage hardware, networking equipment, backup software, and data center infrastructure. These systems also require ongoing maintenance and periodic hardware upgrades.

Cloud storage eliminates many of these expenses by offering a pay-as-you-go pricing model, where organizations pay only for the storage they actually use. Other cost advantages include:

- No upfront hardware investment.
- Automatic storage scaling.
- Reduced maintenance costs.
- Lower electricity and cooling expenses.
- Affordable archival storage options such as Amazon Glacier.
- Reduced administrative overhead.

As business data grows, storage capacity can be increased instantly without purchasing additional hardware.

Disaster Recovery Use Case

Consider a hospital whose primary data center is damaged due to a fire. Patient records, medical reports, and billing information stored on local servers become inaccessible. However, because the hospital has regularly backed up its data to the cloud with cross-region replication enabled, the IT team can quickly restore applications and databases from the cloud backup. This minimizes downtime and allows hospital operations to continue without significant data loss.

Advantages

Cloud-based backup solutions provide several important benefits:

- High durability and reliability.
- Secure data storage.
- Automatic replication.
- Fast disaster recovery.
- Scalable storage capacity.
- Reduced operational costs.
- Business continuity during failures.

Conclusion

Cloud storage offers a highly secure, durable, and cost-effective solution for storing backup data. Features such as redundancy, encryption, access control, automated backups, and disaster recovery ensure that organizations can protect their critical information while reducing infrastructure costs and maintaining continuous business operations.

3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

Introduction

A mobile app company that serves users across different countries must provide fast and reliable access to its application. If all requests are handled from a single server, users who are far from the server may experience high latency, slow loading times, and poor performance. To overcome these challenges, cloud providers use Content Delivery Networks (CDNs) and edge servers to deliver content quickly and efficiently. This improves application performance and ensures a better user experience worldwide.

How Content Delivery Networks (CDNs) Improve Performance

- A Content Delivery Network (CDN) is a network of geographically distributed servers that store cached copies of application content such as images, videos, documents, and other static files. Instead of requesting content from the main server every time, users receive data from the nearest CDN server. This reduces the distance that data must travel, resulting in faster response times and lower latency.
- When a user requests content, the CDN checks whether a cached copy is available at the nearest edge location. If the content is available, it is delivered immediately. If not, the CDN retrieves it from the origin server, stores a copy, and serves it to the user.

The use of CDNs provides several benefits:

- Faster loading of web pages and mobile applications.
- Reduced network latency.
- Lower bandwidth usage on the origin server.
- Reduced server workload.
- Improved application reliability.
- Better performance during high traffic.

Role of Edge Servers

- Edge servers are located close to end users in different geographical regions. These servers store frequently accessed content and process user requests locally instead of forwarding every request to the central data center.
- For example, if a user in India accesses an application whose main server is located in the United States, the request can be served by an edge server located in Mumbai or Chennai. This significantly reduces response time and improves application speed.

Edge servers enhance user experience by:

- Delivering content with minimal delay.
- Reducing buffering during video playback.
- Supporting faster downloads and uploads.
- Decreasing network congestion.
- Improving application responsiveness for users across different regions.

Global Availability Through Cloud Providers

Major cloud providers such as Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP) maintain data centers and CDN edge locations around the world. They provide services that automatically route user requests to the nearest available server using intelligent routing and global load balancing.

Cloud providers support global availability through:

- Multiple geographic regions and availability zones.
- Global Load Balancers for traffic distribution.
- Auto Scaling to handle increasing user demand.
- Multi-region deployment for high availability.
- Automatic failover during server or regional failures.
- Continuous monitoring and health checks.

These features ensure that applications remain available even if one region experiences an outage.

Real-Time Streaming or Gaming Example

- A good example is Netflix, which streams movies and TV shows to millions of users worldwide. Netflix uses CDNs to store video content on edge servers located in different countries. When a user starts watching a movie, the video is streamed from the nearest CDN server instead of a distant central server. This reduces buffering, improves video quality, and provides a smooth viewing experience.
- Similarly, in an online multiplayer game such as PUBG Mobile or Free Fire, edge servers reduce the time taken for player actions to reach the game server. This lowers latency (ping), resulting in smoother gameplay and a better gaming experience.

Advantages of Using CDNs and Edge Servers

The deployment of CDNs and edge servers offers several benefits:

- Faster content delivery.
- Reduced latency.
- Improved application performance.
- Lower load on the origin server.
- Better scalability during peak traffic.
- High availability and reliability.
- Enhanced user satisfaction across different regions.

Conclusion

Content Delivery Networks (CDNs) and edge servers play a vital role in global application deployment. By delivering content from servers located closer to users, they reduce latency, improve application speed, and provide a seamless user experience. Combined with cloud services such as global load balancing, auto-scaling, and multi-region deployment, CDNs ensure that mobile applications remain fast, scalable, and highly available for users around the world.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.

Introduction

Financial institutions such as banks, insurance companies, and investment firms handle highly sensitive information, including customer details, account numbers, transaction records, and payment data. Since this information is confidential, the organization must comply with government regulations and industry standards while ensuring strong security in the cloud. Cloud providers offer various security, compliance, and auditing features to help organizations protect their data and meet legal requirements.

Cloud Providers Support for Regulatory Compliance and Auditing

Major cloud providers such as Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP) comply with international security standards and provide compliance certifications. These certifications help organizations meet legal and regulatory requirements while storing and processing sensitive financial data.

Some commonly supported compliance standards include:

- ISO 27001 – Information Security Management.
- PCI DSS – Security standard for credit and debit card transactions.
- SOC 1, SOC 2, and SOC 3 – Security and operational controls.
- GDPR – Protection of personal data and privacy.
- HIPAA (where applicable) – Protection of sensitive health information.

Cloud providers also offer audit logs and compliance reports, allowing organizations to verify user activities and demonstrate compliance during inspections.

Role of Logging, Monitoring, and Encryption

To maintain cloud security, organizations must continuously monitor their cloud environment.

- Logging records all user activities, including login attempts, file access, configuration changes, and financial transactions. These logs help administrators investigate security incidents and support compliance audits.
- Monitoring continuously checks the health and security of cloud resources. It tracks CPU usage, network traffic, application performance, unauthorized access attempts, and unusual activities. If suspicious behavior is detected, alerts are automatically sent to administrators for immediate action.
- Encryption protects sensitive financial data from unauthorized access.
 - Data at rest should be encrypted using AES-256 encryption.
 - Data in transit should be protected using TLS/SSL encryption.
 - Encryption keys should be securely managed using Key Management Services (KMS).

Together, logging, monitoring, and encryption provide a strong security framework that protects financial information and ensures compliance.

Ensuring Data Sovereignty Requirements

Data sovereignty means that an organization's data must be stored and processed according to the laws of the country where the data belongs. Financial institutions must ensure that customer information does not leave approved geographic locations unless permitted by law.

This can be achieved by:

- Storing data only in approved cloud regions within the required country.
- Using region-specific storage and database services.
- Applying Identity and Access Management (IAM) policies to restrict data access.
- Encrypting sensitive information before storage and transmission.
- Regularly auditing cloud resources to verify compliance with local regulations.

These measures help organizations comply with national data protection laws while maintaining customer trust.

Example: Sensitive Financial Transactions

Consider an online banking system where customers transfer money between accounts. Every transaction is encrypted before being transmitted over the internet using TLS/SSL. The transaction details are stored in encrypted cloud databases using AES-256 encryption. All user actions, such as login attempts, fund transfers, and account updates, are recorded in audit logs. Monitoring services continuously detect unusual activities, such as multiple failed login attempts or suspicious transactions, and immediately alert security administrators. Customer data is stored only in cloud regions that comply with the country's financial regulations, ensuring data sovereignty and regulatory compliance.

Advantages

Implementing cloud security and compliance measures provides several benefits:

- Protects sensitive customer and financial data.
- Ensures compliance with government and industry regulations.
- Prevents unauthorized access through encryption and IAM.
- Supports auditing and investigation using detailed logs.
- Detects security threats through continuous monitoring.
- Builds customer trust by ensuring data privacy and security.
- Helps maintain business continuity and regulatory compliance.

Conclusion

Cloud providers offer comprehensive security and compliance features that enable financial institutions to safely store and process sensitive information. By implementing logging, monitoring, encryption, identity management, and data sovereignty practices, organizations can protect financial transactions, meet regulatory requirements, and provide secure, reliable cloud services to their customers.

5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Introduction

Modern software applications require fast deployment, easy scalability, and efficient resource utilization. Traditional deployment using Virtual Machines (VMs) often consumes more system resources and takes longer to start. Containerization has become a popular solution because it packages an application along with its dependencies, ensuring that it runs consistently across different environments. Tools such as Docker and Kubernetes help organizations deploy, manage, and scale containerized applications efficiently.

Containers vs. Virtual Machines

Containers and Virtual Machines are both used to run applications in isolated environments, but they differ in architecture and resource usage.

A Virtual Machine (VM) includes a complete operating system, application, libraries, and virtual hardware. Since each VM runs its own operating system, it requires more CPU, memory, and storage resources. VMs also take longer to start and occupy more disk space.

In contrast, a Container shares the host operating system kernel while keeping applications isolated. It contains only the application and its required libraries, making it lightweight and efficient. Containers start within seconds and allow many more applications to run on the same physical server compared to VMs.

Comparison

Feature	Containers	Virtual Machines
Operating System	Shares host OS	Separate OS for each VM
Startup Time	Few seconds	Several minutes
Resource Usage	Low	High
Performance	Faster	Slightly slower
Storage Requirement	Small	Large
Portability	High	Moderate

Because of these advantages, containers are widely used in cloud-native application development.

Role of Kubernetes in Deployment and Scaling

When applications consist of many containers, managing them manually becomes difficult. Kubernetes is a container orchestration platform that automates deployment, scaling, networking, and management of containers.

Kubernetes simplifies deployment by automatically scheduling containers on available servers. If a container fails, Kubernetes restarts it automatically, ensuring high availability. It also supports rolling updates, allowing developers to deploy new versions of an application without interrupting users.

For scaling, Kubernetes continuously monitors application workload. During periods of high demand, it automatically creates additional container instances (pods). When demand decreases, extra containers are removed, optimizing resource usage and reducing cloud costs.

Some important Kubernetes features include:

- Automatic deployment of containers.
- Self-healing by restarting failed containers.
- Automatic scaling based on workload.
- Load balancing among containers.
- Rolling updates and rollback support.
- Service discovery for communication between containers.

Portability Across Different Cloud Environments

- One of the biggest advantages of containers is portability. Since containers package the application along with all its dependencies, they can run consistently on different operating systems and cloud platforms without modification.
- For example, a Docker container developed on a local computer can be deployed on Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform (GCP), or even a private cloud without changing the application code. This reduces compatibility issues and prevents vendor lock-in.

Portability provides several benefits:

- Consistent application behavior across environments.
- Easy migration between cloud providers.
- Faster deployment and testing.
- Simplified DevOps and CI/CD pipelines.
- Reduced dependency on a single cloud vendor.

Example of a Microservices-Based Application

A good example is an online shopping (e-commerce) application developed using a microservices architecture. Instead of running as a single large application, it is divided into multiple independent services, each deployed in its own container.

The application may include:

- User Service – Manages user registration and login.
- Product Service – Displays product details.
- Cart Service – Handles shopping cart operations.
- Order Service – Processes customer orders.
- Payment Service – Manages secure online payments.
- Notification Service – Sends email or SMS confirmations.

Each service can be developed, updated, and scaled independently. For example, during a festive sale, only the Product and Order services may need additional containers to handle increased traffic, while the other services continue running normally. Kubernetes automatically manages this scaling process.

Advantages of Containerization

Containerization offers several important benefits:

- Faster application deployment.
- Better resource utilization.
- Lightweight and high-performance execution.
- Easy scalability using Kubernetes.
- High portability across cloud platforms.
- Simplified application management.
- Supports continuous integration and continuous deployment (CI/CD).
- Ideal for microservices architecture.

Conclusion

Containerization has transformed modern application deployment by providing lightweight, portable, and efficient environments for running applications. Compared to Virtual Machines, containers require fewer resources and start much faster. When combined with Kubernetes, organizations can automate deployment, scaling, load balancing, and recovery, making applications more reliable and easier to manage. Therefore, containerization is an ideal solution for developing and deploying modern cloud-native and microservices-based applications.